

Internet (Net)

- Internet stands for International network
- Largest network of the world.
- It is the worldwide collection of computers that links millions of businesses, educational institutions, government organizations, individuals, etc.
- e.g. a computer in Pakistan linked with a computer in England

Applications of Internet

- Education
- E-commerce
- Web applications
- Online payments
- Social media (Facebook, Twitter, LinkedIn, WhatsApp, Instagram, YouTube, TikTok etc.)

Front-end

- HTML
- CSS
- JavaScript
- Node.js

Back-end

- Django
- PHP
- Java
- SQL
- MangoDB

Web Engineering

- Web Engineering: It is the set of methods and tools that allow the development, operations, and maintenance of web-based applications. (now emerged as a subject)
- Web-based applications: web pages, websites, online forms etc.
- Web Engineering consists of theoretical and practical research in Web application development.
- It also handles the navigation structure, security aspects, and requirements of quick changes in previous web applications.

Web Browser: Software used to develop, access & maintain web applications.

- supports contents such as text, audio, video, animation, mathematical formulas etc.
- When we work on web browser, we are working at Front-end.

Web Page & Website

- Web Page: A document which can be displayed in a web browser.
- Website: Collection of a number of web pages

Web Engineer (Web Developer): A person who is responsible to develop & implement a website.

Customer / User / End-user

- A person, a team of people, or an organization, who is the owner of a website.
- E.g. if Meezan Bank asks XYZ software house to develop a website, then Meezan Bank is the Customer/User/End-user of XYZ software house.

Web Servers

- Web Servers: Computers in which webpages, websites & data entered through websites are stored (saved).
- First a website is prepared on a local server (on our own computer), i.e. we work at

Front-end.

- When it is thoroughly checked, then it is made LIVE (Deployment) and stored on a Web Server, so that the whole world can see it. It is stored at Back-end. All the programs and databases are stored at back-end.

Types: Microsoft, Google, Apache, Nginx, lighttpd, and LiteSpeed, each with its own features, advantages, and suitability for different applications.

Link between Web Browser & Web Server

Web Browser

- connected to a Web Server, retrieves and displays the desired document
- supports the display of formatted documents designed in web languages.

World Wide Web (www)

- www: Service provided to locate & access documents (files) on the Internet.
- Other services on the Internet are E-mail, File Transfers, Newsgroups etc.
- supports specially formatted documents in html.

Protocol: Official procedure or system of rules.

Hypertext Transfer Protocol (http)

- Protocol used by the World Wide Web (www) which defines how messages are formatted and transmitted, and what actions web servers and web browsers should take in response to various commands.
- Other protocols used are File Transfer Protocol (FTP), Transmission Control Protocol (TCP), Secure Hypertext Transfer Protocol (https) etc.

Uniform Resource Locator (URL): web address that specifies its location on a computer network and a mechanism for retrieving it.

URL Format protocol://www.domainName.location

Domain Name

- Name of a particular website or webpage
e.g. ssuet.edu
- Consists of an extension called Top-level Domain
e.g. edu in ssuet.edu
- Important Top-level Domains:

edu: Educational institutions e.g. Sir Syed University

gov: Government agencies e.g. NADRA

org: Organizations (non-profitable) e.g. Edhi

mil: Military e.g. Pak Army

com: Commercial/Business e.g. Pak Suzuki

net: Network organizations e.g. PEPSI

Hyperlink (Link)

- It is a digital reference to a web page or website that the user can follow or be guided by clicking or tapping.
- A hyperlink points to a whole document or to a specific portion within a document.

Web Development Process (Requirement Engineering): Steps to develop & Implement a website

Contents of Web Development Process

- Requirement Analysis
- Development Plan
- Site Architecture
- Site Map
- Site Production
- Error checking, Validation & Testing
- Deploying
- Documentation
- Maintenance

Requirement Analysis

- This is the first step to develop a website.
- The Web Developer discusses the requirements of website with the Client (End-user).
- It consists of
 - Feasibility Study
 - Questions to be asked by yourself and end-user

Requirement Analysis - Feasibility Study

- It is a process to examine whether a website should be developed or not
- Includes:
 - **Financial Feasibility:** Are required funds available?
 - **Resource Feasibility:** Are required technology & Manpower available? e.g. computers, servers, Web Engineers, Web/Graphic Designers etc.

Requirement Analysis – Questions to be asked

- Some questions are asked by yourself and others from end-user.
- What are the requirements for the finished site?
e.g. for a car company, models of cars or for an educational institution, admission requirements etc.
- What should be the contents of Home (Main) page & other pages?
- Will your client get full advantage of the website?

- What problems does your client want you to solve? e.g. for car company, new customers to be added or for institution new students to be enrolled
- Who are the target audiences of the website? e.g. for car company, car users or for an educational institution, students
- What is the scope and nature of work? e.g. can we develop similar websites for more car companies or other educational institutions?
- What are the design and programming tasks involved?
- Are there any design tools available, e.g. programming languages, editors etc.
- Who will provide content information for the website, and in what formats? e.g. a Manager in car company or Registrar of a university

Development Plan

- Make a plan
 - Arrange meetings with customers
 - Make schedules e.g. deadlines to make web pages
 - Prepare a list of contents for website
- i.e. textual content, photos (images), audios, videos, logos, copyrights, headers, footers etc.
- Make a list of Functional & Look and Feel requirements

- **Functional Requirements** of a website means what contents a website must show e.g. Garment company's website should show designs of clothing

- **Look & Feel Requirements** should show how those contents look like i.e. includes colors, fonts, images, videos etc.

Site Architecture

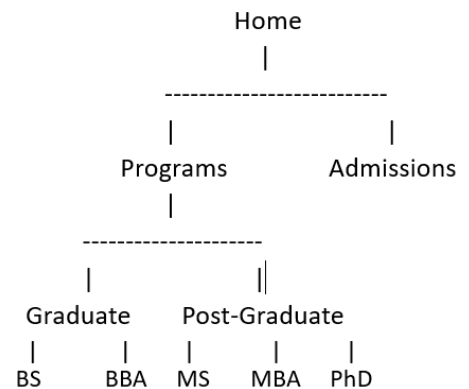
- Design architecture for the site i.e. web page contents etc.
- Architecture depends on nature of information given by: 2 Types of web pages:
- **Static Web Pages** are those whose contents cannot be changed without developer's help e.g. a page containing history of a construction company
- **Dynamic Web Pages**: In which information can be changed by the user after the development e.g. When applying for a job, a candidate fills job application form

Important Features of Website Design

- Information placement: Contents on Home & other web pages
- Layout: Margins etc.
- Visual Effects: Flash banners, change of colors etc.
- Attributes: Fonts, Font Sizes, headings, underline, bold, italics etc.
- Navigation: moving from one page to another

Site Map It is a relationship diagram of all pages to be created, each page having appropriate titles

Example of Site Map of a University Website



Site Production

2 types of Web Development Programming

- Client-side Programming: Programs written and run in a web browser using languages like HTML, JavaScript etc.
- Server-side Programming: Programs written in a web browser to store those programs and databases permanently on web servers. The computer languages used are Django, PHP etc.

Error checking, Validation & Testing

- Once a website is developed, its design and contents are checked thoroughly. Spelling, grammar, navigation, broken links etc. are checked on different browsers
- Different browsers because e.g. if a website is developed in Google Chrome, Mozilla Firefox may show it un-aligned. Therefore for a Web Developer it is important to install atleast 7 to 9 browsers in his/her computer.

Deploying (Web Hosting)

- It is the process of transferring the website from a local computer to a web server.
- A URL (Uniform Resource Locator (address) is issued & the site is registered online.
- Now website can be visited from anywhere in the world i.e. it is LIVE

Documentation: Website Requirements Specification Document

Consists of all the details about the website developed i.e.

- description of the website
- its design and functionalities
- file structure
- User guide etc.

Maintenance: Time-to-time design & content updates by web developers as requested by a user.

HTML Introduction

- HTML stands for Hyper Text Markup Language
- It is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- There are different versions of HTML, latest one is HTML5

<!DOCTYPE>

- <!DOCTYPE> is called declaration
- It represents the document type, and helps browsers to display web pages correctly.
- It must only appear once, at the top of the web page.
- The <!DOCTYPE> declaration for HTML5 is: <!DOCTYPE html>

Elements Other instructions under <!DOCTYPE> are called Elements, i.e. <html>, <head>,

- <html> is called Root Element of html page and shows the start of an html program.
- </html> is also Root Element of html page and shows the end of an html program.
- Between the elements <head> and </head>, the program has Meta Information of the html page for our understanding e.g.

<title>My first web page</title>

It is not displayed on output when we run our program but shown in browser's title bar.

tag : tag is a set of angle brackets that provides instructions to a web browser.

CSS Introduction

- CSS stands for Cascading Style Sheet
- It is the language we use to style a Web page
- CSS describes how HTML elements are to be displayed i.e. it is used to format the layout of a webpage.
- CSS saves a lot of work. It can control the layout of multiple web pages all at once (i.e. we define CSS file only once but it applies to many web pages
- CSS files extension is .css

Internal CSS

- It defines a style for a single HTML page.
- Internal CSS is defined in the <head> section of an HTML page, within a <style> element

External CSS

It is used when you want to define styles for multiple web pages

Advantage of External CSS

- Note that both programs myexternal1.html & myexternal2.html are using the same style sheet mystyle.css
- This is the main advantage of External CSS, you can write a large number of HTML programs which may use one CSS file